

Optimal Noise Schedules in Diffusion Models

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Abstract

We study the model error in discrete-time diffusion models: even with a perfect denoiser, forcing the reverse transitions to be Gaussian introduces error. We derive an exact identity for the averaged one-step Gaussianization error as a function of the noise level, and use it to prove an upper bound that requires no distributional assumptions on the data beyond finite moments from the second step onward. We derive the noise schedule that minimizes the upper bound in closed form, and show that it makes the noise-to-signal odds ratio grow geometrically in the step index.

1 Introduction

Denoising diffusion probabilistic models (DDPMs) [5, 3] generate samples by reversing a noise-adding Markov chain. The forward chain progressively corrupts data with Gaussian noise, and the reverse chain undoes this by applying a learned sequence of Gaussian denoising steps. Even when the denoiser is perfect – that is, when each reverse Gaussian transition is the best possible Gaussian fit to the true reverse conditional – some error remains, because the true reverse conditional is not exactly Gaussian. This irreducible mismatch is the *model error*, and it is the focus of this paper.

We work in the standard DDPM framework. The forward process is a Markov chain $(X_t)_{t=0}^T$ in $\mathbb{R}^d$ defined by

$$X_t = \sqrt{1 - \beta_t} X_{t-1} + \sqrt{\beta_t} Z_t, \quad Z_t \sim \mathcal{N}(0, I_d), \quad (1)$$

where $\beta_1, \dots, \beta_T \in (0, 1]$ are the noise parameters. Writing

$$\bar{\alpha}_t := \prod_{i=1}^t (1 - \beta_i), \quad \bar{\alpha}_0 := 1,$$

we have the explicit representation

$$X_t = \sqrt{\bar{\alpha}_t} X_0 + \sqrt{1 - \bar{\alpha}_t} Z, \quad Z \sim \mathcal{N}(0, I_d).$$

Thus X_t is a progressively Gaussianized version of X_0 .

To sample in reverse, one starts from $\tilde{X}_T \sim \mathcal{N}(0, I_d)$ and generates

$$\tilde{X}_{t-1} \mid \tilde{X}_t \sim \tilde{P}(\tilde{X}_{t-1} \mid \tilde{X}_t),$$

where $\tilde{P}(\cdot \mid \tilde{X}_t)$ is taken to be Gaussian. We assume throughout that each Gaussian reverse kernel is the KL-best Gaussian approximation to the true reverse conditional. The question is then:

How should the schedule $\beta_1, \dots, \beta_T$ be chosen in order to minimize the total model error?

The paper is organized as follows. Section 2 recalls the pathwise KL decomposition that reduces the global model error to a sum of per-step terms. Section 3 develops cumulant identities for Gaussian convolutions that are the technical building blocks of the argument. Section 4, the heart of the paper, derives the exact derivative formula for the averaged one-step error and the resulting upper bound. Section 5 specializes the generic result to the DDPM forward chain. Section 6 solves the schedule optimization problem to obtain the $\beta_1, \dots, \beta_T$ that minimize the upper bound.

2 Setup and pathwise KL decomposition

Let P_t denote the law of X_t under the forward chain (1). For each $t \in \{1, \dots, T\}$, write

$$q_t(\cdot \mid y) := P(X_{t-1} \in \cdot \mid X_t = y)$$

for the true reverse kernel, and let $g_t^*(\cdot \mid y)$ be the Gaussian density minimizing $D_{\text{KL}}(q_t(\cdot \mid y) \parallel g)$ over all Gaussian densities g on $\mathbb{R}^d$. The *Gaussian reverse chain* starts from $\tilde{X}_T \sim \mathcal{N}(0, I_d)$ and applies the kernels g_t^* in reverse; we write $\tilde{P}_0$ for the resulting law at time 0.

Our goal is to bound $D_{\text{KL}}(P_0 \parallel \tilde{P}_0)$, the KL divergence between the true data distribution and the output of the Gaussian reverse chain. However, this quantity is difficult to analyze directly, because the marginal divergence does not decompose into per-step contributions. To obtain such a decomposition, we lift the comparison from marginals to full path measures on $(X_T, \dots, X_0)$:

- the *true reverse law* P_{rev} , with $X_T \sim P_T$ and $X_{t-1} \mid X_t \sim q_t(\cdot \mid X_t)$;
- the *Gaussian reverse law* $\tilde{P}_{\text{rev}}$, with $\tilde{X}_T \sim \mathcal{N}(0, I_d)$ and $\tilde{X}_{t-1} \mid \tilde{X}_t \sim g_t^*(\cdot \mid \tilde{X}_t)$.

Since P_0 and $\tilde{P}_0$ are the time-0 marginals of these two path measures, the data processing inequality [2, Theorem 2.8.1] gives

$$D_{\text{KL}}(P_0 \parallel \tilde{P}_0) \leq D_{\text{KL}}(P_{\text{rev}} \parallel \tilde{P}_{\text{rev}}). \quad (2)$$

The chain rule for KL divergence [2, Theorem 2.5.3] then decomposes the right-hand side into per-step terms:

$$D_{\text{KL}}(P_{\text{rev}} \parallel \tilde{P}_{\text{rev}}) = \underbrace{D_{\text{KL}}(P_T \parallel \mathcal{N}(0, I_d))}_{\text{terminal}} + \sum_{t=1}^T \underbrace{\mathbb{E}[D_{\text{KL}}(q_t(\cdot \mid X_t) \parallel g_t^*(\cdot \mid X_t))]}_{\text{per-step Gaussianization cost}}. \quad (3)$$

The terminal term measures how far the forward chain is from pure noise at time T . Each per-step term is the expected KL cost of replacing the true reverse conditional at step t by its best Gaussian approximation. The remainder of the paper is devoted to bounding these individual terms and optimizing the schedule.

3 Exact identities for Gaussian smoothing

Before turning to the main analysis, we collect two standalone tools about random vectors observed through additive Gaussian noise. These results will be applied in Section 4 to derive the bounds for our diffusion model problem.

The first tool (Lemma 1 and Corollary 1) expresses the derivatives of the log-density of the observation in terms of conditional cumulants of the noise. The second tool (Proposition 1) bounds the averaged squared Frobenius norm of the third derivative.

3.1 A conditional mgf identity

Let V be an arbitrary random vector in $\mathbb{R}^d$, let $Z \sim \mathcal{N}(0, I_d)$ be independent of V , fix $\sigma > 0$, and define

$$Y := V + \sigma Z.$$

Since Y is a Gaussian convolution, its density $p(y) = \mathbb{E}[\phi_\sigma(y - V)]$ (where ϕ_σ is the density of $\mathcal{N}(0, \sigma^2 I_d)$) is everywhere positive and C^∞ . In particular, $\ell := \log p$ is a well-defined smooth function on all of $\mathbb{R}^d$.

Lemma 1 (Conditional mgf of the Gaussian noise). *For every $u \in \mathbb{R}^d$ and every $y \in \mathbb{R}^d$,*

$$\mathbb{E}[e^{u^\top Z} \mid Y = y] = \exp\left(\frac{1}{2}\|u\|^2\right) \frac{p(y - \sigma u)}{p(y)}. \quad (4)$$

Consequently,

$$\mathbb{E}[Z \mid Y = y] = -\sigma \nabla \ell(y), \quad (5)$$

$$\text{Cov}(Z \mid Y = y) = I_d + \sigma^2 \nabla^2 \ell(y), \quad (6)$$

$$\kappa_3(Z \mid Y = y) = -\sigma^3 \nabla^3 \ell(y). \quad (7)$$

Here $\kappa_3(Z \mid Y = y)$ is the conditional third cumulant tensor, defined as the third-order tensor with entries

$$\kappa_3(Z \mid Y = y)_{ijk} := \mathbb{E}[(Z_i - \mu_i)(Z_j - \mu_j)(Z_k - \mu_k) \mid Y = y], \quad \mu := \mathbb{E}[Z \mid Y = y].$$

Proof. We will show that $\mathbb{E}[e^{u^\top Z} f(Y)] = \mathbb{E}[g_u(Y) f(Y)]$ for every bounded Borel function f , where $g_u(y) := e^{\|u\|^2/2} p(y - \sigma u)/p(y)$. Since a random variable is determined by its expectation against all bounded Borel functions of Y , this identifies $g_u(Y)$ as $\mathbb{E}[e^{u^\top Z} \mid Y]$.

Fix $u \in \mathbb{R}^d$ and let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be bounded and Borel. Using the Gaussian identity $e^{u^\top z} \phi(z) = e^{\|u\|^2/2} \phi(z - u)$ and substituting $w = z - u$,

$$\begin{aligned} \mathbb{E}[e^{u^\top Z} f(Y)] &= \mathbb{E}\left[\int_{\mathbb{R}^d} e^{u^\top z} f(V + \sigma z) \phi(z) dz\right] \\ &= e^{\|u\|^2/2} \mathbb{E}\left[\int_{\mathbb{R}^d} f(V + \sigma w + \sigma u) \phi(w) dw\right] = e^{\|u\|^2/2} \mathbb{E}[f(Y + \sigma u)]. \end{aligned}$$

Since Y has density p , the right-hand side equals $e^{\|u\|^2/2} \int f(y) p(y - \sigma u) dy$. Therefore

$$\mathbb{E}[e^{u^\top Z} f(Y)] = \int f(y) e^{\|u\|^2/2} p(y - \sigma u) dy$$

for every bounded Borel f . Writing $g_u(y) := e^{\|u\|^2/2} p(y - \sigma u)/p(y)$ (which is well-defined since $p > 0$ everywhere), the right-hand side equals $\int f(y) g_u(y) p(y) dy = \mathbb{E}[g_u(Y) f(Y)]$. Since

$$\mathbb{E}[e^{u^\top Z} f(Y)] = \mathbb{E}[g_u(Y) f(Y)]$$

holds for every bounded Borel f , we conclude $\mathbb{E}[e^{u^\top Z} \mid Y = y] = g_u(y)$ for every y , proving (4).

Taking logarithms gives the conditional cumulant generating function

$$\log \mathbb{E}[e^{u^\top Z} \mid Y = y] = \frac{1}{2}\|u\|^2 + \ell(y - \sigma u) - \ell(y).$$

Differentiating at $u = 0$ yields (5)–(7). □

Since $V = Y - \sigma Z$ given Y , the preceding lemma immediately yields the posterior mean and covariance of V .

Corollary 1 (Posterior mean and covariance). *With the notation above,*

$$\begin{aligned}\mathbb{E}[V \mid Y = y] &= y + \sigma^2 \nabla \ell(y), \\ \text{Cov}(V \mid Y = y) &= \sigma^2 (I_d + \sigma^2 \nabla^2 \ell(y)).\end{aligned}$$

Moreover,

$$\kappa_3(V \mid Y = y) = \sigma^6 \nabla^3 \ell(y). \quad (8)$$

Proof. Since $V = y - \sigma Z$ given $Y = y$, we have $\mathbb{E}[V \mid Y = y] = y - \sigma \mathbb{E}[Z \mid Y = y]$, $\text{Cov}(V \mid Y = y) = \sigma^2 \text{Cov}(Z \mid Y = y)$, and $\kappa_3(V \mid Y = y) = (-\sigma)^3 \kappa_3(Z \mid Y = y)$. Substituting (5)–(7) gives the result. $\square$

3.2 Averaged bounds for the third derivative of $\log p$

We write $\|\cdot\|_F$ for the Frobenius norm of a matrix or tensor. For a third-order tensor $T = (T_{ijk})$, this means

$$\|T\|_F^2 := \sum_{i,j,k=1}^d T_{ijk}^2.$$

Proposition 1 (Averaged third-derivative bounds for a Gaussian convolution). *Let $Y = V + \sigma Z$ with $\sigma > 0$ and $Z \sim \mathcal{N}(0, I_d)$ independent of V . Let p be the density of Y , and write $\ell = \log p$. Then*

$$\mathbb{E}[\|\nabla^3 \ell(Y)\|_F^2] \leq \frac{64 d(d+2)(d+4)}{\sigma^6}. \quad (9)$$

If, in addition, $\mathbb{E}\|V\|^6 < \infty$, then

$$\mathbb{E}[\|\nabla^3 \ell(Y)\|_F^2] \leq \frac{64 \mathbb{E}\|V\|^6}{\sigma^{12}}. \quad (10)$$

Proof. By (7),

$$\sigma^3 \nabla^3 \ell(Y) = -\kappa_3(Z \mid Y).$$

Since third cumulants are translation-invariant, $\kappa_3(Z \mid Y)$ equals the third moment tensor of $Z - \mathbb{E}[Z \mid Y]$ given Y :

$$\kappa_3(Z \mid Y) = \mathbb{E}[(Z - \mathbb{E}[Z \mid Y])^{\otimes 3} \mid Y].$$

Applying Jensen's inequality (with the convex function $\|\cdot\|_F^2$) to the conditional expectation, and using the identity $\|x^{\otimes 3}\|_F^2 = \|x\|^6$, gives

$$\|\kappa_3(Z \mid Y)\|_F^2 \leq \mathbb{E}[\|(Z - \mathbb{E}[Z \mid Y])^{\otimes 3}\|_F^2 \mid Y] = \mathbb{E}[\|Z - \mathbb{E}[Z \mid Y]\|^6 \mid Y].$$

Taking expectations and using the triangle inequality followed by $(a+b)^6 \leq 32(a^6 + b^6)$,

$$\begin{aligned}\mathbb{E}[\|\kappa_3(Z \mid Y)\|_F^2] &\leq \mathbb{E}\|Z - \mathbb{E}[Z \mid Y]\|^6 \\ &\leq 32 \mathbb{E}\|Z\|^6 + 32 \mathbb{E}\|\mathbb{E}[Z \mid Y]\|^6.\end{aligned}$$

For the second term, Jensen's inequality gives $\|\mathbb{E}[Z \mid Y]\|^6 \leq \mathbb{E}\|Z\|^6 \mid Y$, so $\mathbb{E}\|\mathbb{E}[Z \mid Y]\|^6 \leq \mathbb{E}\|Z\|^6$. Therefore

$$\mathbb{E}[\|\kappa_3(Z \mid Y)\|_F^2] \leq 64 \mathbb{E}\|Z\|^6 = 64 d(d+2)(d+4).$$

Dividing by σ^6 proves (9).

For the refined bound, use (8) instead:

$$\sigma^6 \nabla^3 \ell(Y) = \kappa_3(V \mid Y).$$

Exactly the same argument gives

$$\mathbb{E}[\|\kappa_3(V \mid Y)\|_F^2] \leq 64 \mathbb{E}\|V\|^6,$$

which yields (10) after dividing by σ^{12} . $\square$

4 An exact formula for the averaged one-step error

The identities of the previous section describe individual derivatives of the log-density of a Gaussian convolution. We now combine them to analyze the quantity that appears in the KL decomposition (3): the expected KL divergence between the exact posterior and its best Gaussian approximation.

The strategy is to view this averaged error as a function of the noise level, derive an exact formula for its derivative, and then integrate. The derivative turns out to involve only the third spatial derivatives of the current log-density, which the bounds from Proposition 1 control.

4.1 A generic one-step setup

We work in a generic observation model that will later be matched to each step of the diffusion chain. Let U be a random vector in $\mathbb{R}^d$ with $\mathbb{E}\|U\|^2 < \infty$, let $W, Z \sim \mathcal{N}(0, I_d)$ be independent of each other and of U , and fix $\eta > 0$. Define

$$S := U + \sqrt{\eta} W, \quad Y_u := S + \sqrt{u} Z, \quad u \geq 0. \quad (11)$$

For every $u \geq 0$, let p_u be the density of Y_u , and write

$$\ell_u := \log p_u, \quad H_u := \nabla^2 \ell_u, \quad A_u := I_d + u H_u.$$

Throughout, $\partial_r := \partial/\partial y_r$ denotes the partial derivative in the r -th coordinate of y , while ∂_u denotes the derivative in the noise level u . All gradients, Hessians, and Laplacians (∇, ∇^2, Δ) are taken in the spatial variable y . For each $y \in \mathbb{R}^d$, let

$$q_u(\cdot \mid y) := \mathcal{L}(S \mid Y_u = y)$$

be the exact posterior law.

Lemma 2 (The KL-best Gaussian is the moment-matched Gaussian). *Let Q be any probability measure on $\mathbb{R}^d$ with finite second moment. Among all Gaussian distributions g on $\mathbb{R}^d$, the quantity $D_{\text{KL}}(Q \parallel g)$ is minimized by the Gaussian with the same mean and covariance as Q .*

Proof. Write $G_\star := N(m_Q, \Sigma_Q)$ for the moment-matched Gaussian (which may be degenerate) and let $A := m_Q + \text{Ran}(\Sigma_Q)$. First, $Q(A) = 1$: if $v \in \ker(\Sigma_Q)$, then $\text{Var}(v^\top X) = v^\top \Sigma_Q v = 0$, so $v^\top X = v^\top m_Q$ a.s., which forces $X - m_Q \in \text{Ran}(\Sigma_Q)$ a.s. Moreover, A is the smallest affine subspace carrying full Q -mass.

Let G be any Gaussian measure on $\mathbb{R}^d$. If $D_{\text{KL}}(Q \parallel G) = +\infty$ there is nothing to prove. Otherwise $Q \ll G$. Let B be the affine support of G , so $G(B) = 1$ and hence $Q(B) = 1$. By

minimality of A , we have $A \subseteq B$. If $A \subsetneq B$, then A is a proper affine subspace of B , so $G(A) = 0$, contradicting $Q \ll G$ and $Q(A) = 1$. Therefore $B = A$.

Thus both G and $G_\star$ have strictly positive densities g and $g_\star$ with respect to Lebesgue measure λ_A on A . Since $Q \ll G \ll \lambda_A$, write $q := dQ/d\lambda_A$. Now $\log(g_\star/g)$ is a polynomial of degree at most two on A , and Q and $G_\star$ share the same mean and covariance, so

$$\int_A \log \frac{g_\star}{g} dQ = \int_A \log \frac{g_\star}{g} dG_\star = D_{\text{KL}}(G_\star \parallel G).$$

Hence

$$D_{\text{KL}}(Q \parallel G) = \int_A \log \frac{q}{g_\star} dQ + \int_A \log \frac{g_\star}{g} dQ = D_{\text{KL}}(Q \parallel G_\star) + D_{\text{KL}}(G_\star \parallel G) \geq D_{\text{KL}}(Q \parallel G_\star).$$

If $D_{\text{KL}}(Q \parallel G_\star) < \infty$, equality holds if and only if $G = G_\star$. $\square$

By Lemma 2, the optimal Gaussian approximation to $q_u(\cdot \mid y)$ is simply the Gaussian with mean $\mathbb{E}[S \mid Y_u = y]$ and covariance $\text{Cov}(S \mid Y_u = y)$.

Corollary 2 (Posterior mean and covariance in the generic setup). *For the setup (11), the KL-best Gaussian posterior is*

$$g_u^\star(\cdot \mid y) = N(m_u(y), \Sigma_u(y)),$$

where

$$m_u(y) = y + u \nabla \ell_u(y), \tag{12}$$

$$\Sigma_u(y) = u A_u(y) = u(I_d + u \nabla^2 \ell_u(y)). \tag{13}$$

Moreover,

$$A_u(y) \succeq 0 \quad \text{for all } u > 0, y \in \mathbb{R}^d.$$

Proof. Apply Corollary 1 with $V = S$, $\sigma = \sqrt{u}$, and $Y = Y_u$. This gives (12) and (13). Since $\Sigma_u(y)$ is a covariance matrix, it is positive semidefinite, and therefore so is $A_u(y) = u^{-1} \Sigma_u(y)$. $\square$

Define the averaged one-step Gaussianization error

$$\mathcal{K}(u) := \mathbb{E}_{Y_u} \left[D_{\text{KL}}(q_u(\cdot \mid Y_u) \parallel g_u^\star(\cdot \mid Y_u)) \right]. \tag{14}$$

4.2 An entropy representation

The first formula is purely information-theoretic.

Proposition 2 (Entropy representation of $\mathcal{K}(u)$). *For every $u > 0$,*

$$\mathcal{K}(u) = h(Y_u) - h(S) + \frac{1}{2} \mathbb{E}[\log \det A_u(Y_u)]. \tag{15}$$

Proof. For a Gaussian density g , $\log g(x)$ is a quadratic polynomial in x , so $\int \log g dQ$ depends on Q only through its mean and covariance. When G is the moment-matched Gaussian (same mean and covariance as Q), this gives $\int \log g dQ = \int \log g dG = -h(G)$, and therefore $D_{\text{KL}}(Q \parallel G) = -h(Q) + h(G)$. Applying this pointwise with $Q = q_u(\cdot \mid y)$ and $G = g_u^\star(\cdot \mid y)$:

$$D_{\text{KL}}(q_u(\cdot \mid y) \parallel g_u^\star(\cdot \mid y)) = h(g_u^\star(\cdot \mid y)) - h(q_u(\cdot \mid y)).$$

Average over Y_u . The Gaussian entropy formula gives

$$h(g_u^\star(\cdot | y)) = \frac{1}{2} \log \left((2\pi e)^d \det \Sigma_u(y) \right).$$

Using (13),

$$\det \Sigma_u(y) = u^d \det A_u(y).$$

Therefore

$$\mathbb{E}[h(g_u^\star(\cdot | Y_u))] = \frac{d}{2} \log(2\pi e u) + \frac{1}{2} \mathbb{E}[\log \det A_u(Y_u)].$$

On the other hand, by the definition of conditional entropy,

$$\mathbb{E}[h(q_u(\cdot | Y_u))] = h(S | Y_u).$$

Finally, applying the chain rule for entropy gives

$$h(S | Y_u) = h(S) + h(Y_u | S) - h(Y_u),$$

and because $Y_u = S + \sqrt{u}Z$,

$$h(Y_u | S) = h(\sqrt{u}Z) = \frac{d}{2} \log(2\pi e u).$$

Substituting these identities into the definition of $\mathcal{K}(u)$ proves (15). $\square$

Lemma 3 (Right-continuity of $\mathcal{K}$ at the origin). *If $\mathbb{E}\|U\|^2 < \infty$, then $\mathcal{K}(u) \rightarrow 0$ as $u \rightarrow 0^+$.*

Proof. By the entropy representation,

$$\mathcal{K}(u) = [h(Y_u) - h(S)] + \frac{1}{2} \mathbb{E}[\log \det A_u(Y_u)],$$

and we show that each term vanishes as $u \rightarrow 0^+$.

Entropy difference. By de Bruijn's identity, $h(Y_u) - h(S) = \frac{1}{2} \int_0^u J(Y_s) ds$, where $J(Y_s) = \mathbb{E}\|\nabla \ell_s(Y_s)\|^2$ is the Fisher information. Since Y_s has total Gaussian smoothing $\eta + s \geq \eta > 0$, Stam's inequality applied to the Gaussian component yields $J(Y_s) \leq d/(\eta + s) \leq d/\eta$. Hence $|h(Y_u) - h(S)| \leq du/(2\eta) \rightarrow 0$.

Log-determinant term. The lower bound $A_u(y) \succeq \frac{\eta}{\eta+u} I_d$ gives $\log \det A_u(y) \geq -d \log(1+u/\eta) \rightarrow 0$ uniformly in y . For the upper bound, diagonalizing $I_d + uH_u(y)$ and applying $\log(1+x) \leq x$ to each eigenvalue (each of which is positive, since $A_u \succ 0$) gives the pointwise inequality $\log \det A_u(y) \leq u \operatorname{tr} H_u(y) = u \Delta \ell_u(y)$. Taking expectations,

$$\mathbb{E}[\log \det A_u(Y_u)] \leq u \mathbb{E}[\Delta \ell_u(Y_u)] = u \int (\Delta \ell_u) p_u dy = -u \int \|\nabla \ell_u\|^2 p_u dy = -u J(Y_u) \leq 0,$$

where the middle equality is the standard integration by parts $\int (\Delta \ell_u) p_u = -\int \nabla \ell_u \cdot \nabla p_u = -\int \|\nabla \ell_u\|^2 p_u$ (valid because p_u is a Gaussian convolution, so it and its gradient vanish at infinity and the boundary terms drop). Combining the two bounds, $-d \log(1+u/\eta) \leq \mathbb{E}[\log \det A_u(Y_u)] \leq 0$, so $\mathbb{E}[\log \det A_u(Y_u)] \rightarrow 0$. $\square$

4.3 The exact derivative formula

The next theorem is the central identity of the paper.

Theorem 1 (Exact derivative formula for averaged Gaussianization). *For every $u > 0$, the function $\mathcal{K}(u)$ is differentiable and*

$$\mathcal{K}'(u) = \frac{1}{4} \mathbb{E} \left[\sum_{r=1}^d \|A_u(Y_u)^{-1/2} (\partial_r A_u)(Y_u) A_u(Y_u)^{-1/2}\|_F^2 \right]. \quad (16)$$

Equivalently, writing $T_{u,r} := \partial_r H_u$ for the r -th spatial derivative of the Hessian,

$$\mathcal{K}'(u) = \frac{u^2}{4} \mathbb{E} \left[\sum_{r=1}^d \|A_u(Y_u)^{-1/2} T_{u,r}(Y_u) A_u(Y_u)^{-1/2}\|_F^2 \right]. \quad (17)$$

In particular, $\mathcal{K}'(u) \geq 0$.

Remark 1 (Why differentiate?). *Proposition 2 gives an exact formula for $\mathcal{K}(u)$, but its individual terms are difficult to bound directly. Differentiating yields a cleaner structure: the derivative $\mathcal{K}'(u)$ depends only on the third-order residual, which is precisely what Proposition 1 controls. Integrating from $\mathcal{K}(0) = 0$ then gives the bound on $\mathcal{K}(u)$.*

Proof. We differentiate the entropy representation (15) term by term. We first compute the entropy derivative via de Bruijn's identity. We then compute the log det derivative using the heat equation and integration by parts.

Fix $u > 0$. Since Y_u is a Gaussian convolution with total Gaussian variance $\eta + u$, the density $p_u(y)$ is positive and C^∞ jointly in (u, y) , and so are $\ell_u = \log p_u$, the Hessian $H_u = \nabla^2 \ell_u$, and $A_u = I_d + u H_u$. Moreover, $A_u(y) \succeq \frac{\eta}{\eta+u} I_d \succ 0$ for all y (by (6) applied with $\sigma = \sqrt{\eta+u}$), so $\log \det A_u(y)$ is well-defined and C^∞ jointly in (u, y) .

Differentiate the entropy term. Our goal here is to show that

$$\frac{d}{du} h(Y_u) = -\frac{1}{2} \mathbb{E}[\text{tr} H_u(Y_u)].$$

By de Bruijn's identity [6],

$$\frac{d}{du} h(Y_u) = \frac{1}{2} \mathbb{E}[\|\nabla \ell_u(Y_u)\|^2].$$

Using the fact that $\nabla \ell_u(y) = \nabla p_u(y)/p_u(y)$, we have

$$\mathbb{E}[\|\nabla \ell_u(Y_u)\|^2] = \int \|\nabla \ell_u(x)\|^2 p_u(x) dx = \int \nabla \ell_u(x) \cdot \nabla p_u(x) dx.$$

Using integration by parts, we get

$$\int \nabla \ell_u(x) \cdot \nabla p_u(x) dx = - \int \left(\sum_r \frac{\partial^2 \ell_u(x)}{\partial x_r^2} \right) p_u(x) dx = -\mathbb{E}[\text{tr}(H_u(Y_u))].$$

Thus, $\mathbb{E}[\|\nabla \ell_u(Y_u)\|^2] = -\mathbb{E}[\text{tr}(H_u(Y_u))]$, so the entropy derivative is

$$\frac{d}{du} h(Y_u) = -\frac{1}{2} \mathbb{E}[\text{tr} H_u(Y_u)]. \quad (18)$$

Differentiate $\mathbb{E}[\log \det A_u(Y_u)]$. Set

$$\Phi(u) := \mathbb{E}[\log \det A_u(Y_u)].$$

Writing $\Delta := \sum_{i=1}^d \partial_{ii}$ for the Laplacian, we apply Lemma 5 (see Appendix A) with

$$f_u(y) = \log \det A_u(y),$$

which gives

$$\Phi'(u) = \mathbb{E}[\partial_u \log \det A_u(Y_u)] + \frac{1}{2} \mathbb{E}[\Delta \log \det A_u(Y_u)]. \quad (19)$$

The polynomial-growth hypothesis of Lemma 5 holds for $f_u = \log \det A_u$: the uniform bound $A_u \succeq \frac{\eta}{\eta+u} I_d$ gives a positive lower bound on $\det A_u$, so $\log \det A_u$ is bounded below and grows at most logarithmically in y ; the spatial and u -derivatives of $\log \det A_u$ are rational functions of entries of A_u^{-1} and of spatial derivatives of $H_u = \nabla^2 \ell_u$, each of which has polynomial growth in y under the moment assumptions on U . Then

$$\partial_u A_u = H_u + u \partial_u H_u.$$

Hence

$$\partial_u \log \det A_u = \text{tr}(A_u^{-1} H_u) + u \text{tr}(A_u^{-1} \partial_u H_u). \quad (20)$$

Since $Y_u = S + \sqrt{u}Z$, its density p_u satisfies the heat equation $\partial_u p_u = \frac{1}{2} \Delta p_u$. Rewriting this equation in terms of the log-density $\ell_u = \log p_u$, we get

$$\partial_u \ell_u = \frac{1}{2} \Delta \ell_u + \frac{1}{2} \|\nabla \ell_u\|^2.$$

Differentiating twice in y gives, for each pair (i, j) ,

$$\partial_u (H_u)_{ij} = \frac{1}{2} \Delta (H_u)_{ij} + \frac{1}{2} \partial_{ij} \left[\sum_{r=1}^d (\partial_r \ell_u)^2 \right].$$

Expanding the last term by the product rule,

$$\frac{1}{2} \partial_{ij} \left[\sum_{r=1}^d (\partial_r \ell_u)^2 \right] = \partial_j \left[\sum_{r=1}^d (\partial_r \ell_u) (\partial_{ir} \ell_u) \right] = \sum_{r=1}^d (\partial_r \ell_u) (\partial_{ijr} \ell_u) + \sum_{r=1}^d (H_u)_{jr} (H_u)_{ir},$$

Hence

$$\partial_u (H_u)_{ij} = \frac{1}{2} \Delta (H_u)_{ij} + \sum_{r=1}^d (\partial_r \ell_u) (\partial_{ijr} \ell_u) + \sum_{r=1}^d (H_u)_{ir} (H_u)_{jr}.$$

In matrix notation, writing $T_{u,r} := \partial_r H_u$ for the matrix of r -th spatial derivatives of H_u ,

$$\partial_u H_u = \frac{1}{2} \Delta H_u + \sum_{r=1}^d (\partial_r \ell_u) T_{u,r} + H_u^2. \quad (21)$$

Substitute (21) into (20):

$$\begin{aligned} \partial_u \log \det A_u &= \text{tr}(A_u^{-1} H_u) + \frac{u}{2} \text{tr}(A_u^{-1} \Delta H_u) \\ &\quad + u \sum_{r=1}^d (\partial_r \ell_u) \text{tr}(A_u^{-1} T_{u,r}) + u \text{tr}(A_u^{-1} H_u^2). \end{aligned} \quad (22)$$

Next, compute the Laplacian term in (19). Since $\partial_r A_u = uT_{u,r}$,

$$\partial_r \log \det A_u = u \operatorname{tr}(A_u^{-1} T_{u,r}).$$

Differentiate once more. Using

$$\partial_r(A_u^{-1}) = -A_u^{-1}(\partial_r A_u)A_u^{-1} = -uA_u^{-1}T_{u,r}A_u^{-1},$$

we get

$$\partial_{rr} \log \det A_u = u \operatorname{tr}(A_u^{-1} \partial_r T_{u,r}) - u^2 \operatorname{tr}(A_u^{-1} T_{u,r} A_u^{-1} T_{u,r}).$$

Summing over r ,

$$\Delta \log \det A_u = u \operatorname{tr}(A_u^{-1} \Delta H_u) - u^2 \sum_{r=1}^d \operatorname{tr}(A_u^{-1} T_{u,r} A_u^{-1} T_{u,r}). \quad (23)$$

Now substitute (22) and (23) into (19). This gives

$$\begin{aligned} \Phi'(u) &= \mathbb{E}[\operatorname{tr}(A_u^{-1} H_u)] + u \mathbb{E}[\operatorname{tr}(A_u^{-1} H_u^2)] \\ &\quad + u \sum_{r=1}^d \mathbb{E}[(\partial_r \ell_u) \operatorname{tr}(A_u^{-1} T_{u,r})] \\ &\quad + u \mathbb{E}[\operatorname{tr}(A_u^{-1} \Delta H_u)] - \frac{u^2}{2} \mathbb{E}\left[\sum_{r=1}^d \operatorname{tr}(A_u^{-1} T_{u,r} A_u^{-1} T_{u,r})\right]. \end{aligned} \quad (24)$$

The third and fourth terms simplify by integration by parts. For any sufficiently smooth scalar function f ,

$$\mathbb{E}[(\partial_r \ell_u)(Y_u) f(Y_u)] = \int (\partial_r p_u) f = - \int p_u (\partial_r f) = -\mathbb{E}[(\partial_r f)(Y_u)],$$

where the middle equality is the same integration-by-parts argument used in the proof of Lemma 5 (see Appendix A). Apply this with $f(y) = \operatorname{tr}(A_u^{-1}(y) T_{u,r}(y))$. Since

$$\partial_r f = \operatorname{tr}(A_u^{-1} \partial_r T_{u,r}) - u \operatorname{tr}(A_u^{-1} T_{u,r} A_u^{-1} T_{u,r}),$$

we obtain

$$\mathbb{E}[(\partial_r \ell_u) \operatorname{tr}(A_u^{-1} T_{u,r})] = -\mathbb{E}[\operatorname{tr}(A_u^{-1} \partial_r T_{u,r})] + u \mathbb{E}[\operatorname{tr}(A_u^{-1} T_{u,r} A_u^{-1} T_{u,r})].$$

Summing over r , and using $\sum_r \partial_r T_{u,r} = \Delta H_u$, the third and fourth terms in (24) combine to

$$-u \mathbb{E}[\operatorname{tr}(A_u^{-1} \Delta H_u)] + u^2 \mathbb{E}\left[\sum_{r=1}^d \operatorname{tr}(A_u^{-1} T_{u,r} A_u^{-1} T_{u,r})\right] + u \mathbb{E}[\operatorname{tr}(A_u^{-1} \Delta H_u)],$$

so the ΔH_u terms cancel exactly. Therefore

$$\Phi'(u) = \mathbb{E}[\operatorname{tr}(A_u^{-1} H_u)] + u \mathbb{E}[\operatorname{tr}(A_u^{-1} H_u^2)] + \frac{u^2}{2} \mathbb{E}\left[\sum_{r=1}^d \operatorname{tr}(A_u^{-1} T_{u,r} A_u^{-1} T_{u,r})\right]. \quad (25)$$

Factoring H_u on the right, $H_u + uH_u^2 = (I_d + uH_u)H_u = A_u H_u$, and therefore

$$A_u^{-1} H_u + uA_u^{-1} H_u^2 = A_u^{-1} (H_u + uH_u^2) = A_u^{-1} A_u H_u = H_u.$$

Taking traces and expectations, we arrive at

$$\Phi'(u) = \mathbb{E}[\operatorname{tr} H_u(Y_u)] + \frac{u^2}{2} \mathbb{E}\left[\sum_{r=1}^d \operatorname{tr}(A_u^{-1} T_{u,r} A_u^{-1} T_{u,r})\right]. \quad (26)$$

Combine with the entropy derivative. Differentiate (15). Using (18) and (26),

$$\begin{aligned}\mathcal{K}'(u) &= \frac{d}{du} h(Y_u) + \frac{1}{2} \Phi'(u) \\ &= -\frac{1}{2} \mathbb{E}[\text{tr} H_u(Y_u)] + \frac{1}{2} \mathbb{E}[\text{tr} H_u(Y_u)] + \frac{u^2}{4} \mathbb{E} \left[\sum_{r=1}^d \text{tr}(A_u^{-1} T_{u,r} A_u^{-1} T_{u,r}) \right].\end{aligned}$$

This matches (17) after rewriting the trace as a Frobenius norm: for symmetric T and positive-definite A , cyclicity gives

$$\text{tr}(A^{-1} T A^{-1} T) = \text{tr}((A^{-1/2} T A^{-1/2})^2) = \|A^{-1/2} T A^{-1/2}\|_F^2,$$

applied here with $T = T_{u,r}$ (symmetric, since H_u is) and $A = A_u$. Since $\partial_r A_u = u T_{u,r}$, (16) follows as well. The nonnegativity of $\mathcal{K}'(u)$ is automatic from the squared-norm form. $\square$

4.4 A universal bound from pre-existing smoothing

The derivative formula becomes useful as soon as S already contains some Gaussian smoothing.

Theorem 2 (One-step Gaussianization bound with pre-existing smoothing). *Assume the setup (11). For every $u \geq 0$,*

$$\mathcal{K}(u) \leq 16 d(d+2)(d+4) \Psi\left(\frac{u}{\eta}\right), \quad \Psi(r) := \frac{r^2}{2} - r + \log(1+r). \quad (27)$$

Moreover, if $\mathbb{E}\|U\|^6 < \infty$, then

$$\mathcal{K}(u) \leq \frac{16}{3} \mathbb{E}\|U\|^6 \frac{u^3}{\eta^3(\eta+u)^3}. \quad (28)$$

Finally, for every $r \in [0, 1]$,

$$\Psi(r) \leq \frac{r^3}{3}. \quad (29)$$

Therefore, if $u \leq \eta$, then

$$\mathcal{K}(u) \leq \frac{16}{3} d(d+2)(d+4) \left(\frac{u}{\eta}\right)^3. \quad (30)$$

Proof. Recall from Theorem 1, equation (17), that

$$\mathcal{K}'(u) = \frac{u^2}{4} \mathbb{E} \left[\sum_{r=1}^d \|A_u(Y_u)^{-1/2} T_{u,r}(Y_u) A_u(Y_u)^{-1/2}\|_F^2 \right].$$

We bound the inner Frobenius norm pointwise in y . For any two matrices M and B , we have $\|BMB\|_F \leq \|B\|_{\text{op}}^2 \|M\|_F$. Applying this with $M = T_{u,r}(y)$ and $B = A_u(y)^{-1/2}$, and using $\|A_u(y)^{-1/2}\|_{\text{op}}^2 = \|A_u(y)^{-1}\|_{\text{op}}$,

$$\|A_u(y)^{-1/2} T_{u,r}(y) A_u(y)^{-1/2}\|_F^2 \leq \|A_u(y)^{-1}\|_{\text{op}}^2 \|T_{u,r}(y)\|_F^2.$$

Summing over r and using that $(T_{u,r})_{ij} = \partial_r \partial_i \partial_j \ell_u$ is the r -th slice of $\nabla^3 \ell_u$, so $\sum_r \|T_{u,r}(y)\|_F^2 = \|\nabla^3 \ell_u(y)\|_F^2$, we obtain

$$\mathcal{K}'(u) \leq \frac{u^2}{4} \mathbb{E} \left[\|A_u(Y_u)^{-1}\|_{\text{op}}^2 \|\nabla^3 \ell_u(Y_u)\|_F^2 \right]. \quad (31)$$

To proceed, we derive a uniform upper bound on $\|A_u(y)^{-1}\|_{\text{op}}$. Since

$$Y_u = U + \sqrt{\eta + u} \tilde{Z} \quad \text{for } \tilde{Z} \sim \mathcal{N}(0, I_d),$$

Equation (6) applied with $V = U$ and $\sigma = \sqrt{\eta + u}$ gives

$$I_d + (\eta + u)H_u(y) = \text{Cov}(\tilde{Z} \mid Y_u = y) \succeq 0, \quad \text{so} \quad H_u(y) \succeq -\frac{1}{\eta + u}I_d.$$

Therefore

$$A_u(y) = I_d + uH_u(y) \succeq \frac{\eta}{\eta + u}I_d, \quad \|A_u(y)^{-1}\|_{\text{op}} \leq \frac{\eta + u}{\eta} \quad \text{for all } y \in \mathbb{R}^d.$$

Substituting this uniform bound into (31),

$$\mathcal{K}'(u) \leq \frac{u^2}{4} \left(\frac{\eta + u}{\eta} \right)^2 \mathbb{E}[\|\nabla^3 \ell_u(Y_u)\|_{\text{F}}^2]. \quad (32)$$

We now use Proposition 1. Since $Y_u = U + \sqrt{\eta + u} \tilde{Z}$, the universal bound (9) gives

$$\mathbb{E}[\|\nabla^3 \ell_u(Y_u)\|_{\text{F}}^2] \leq \frac{64 d(d+2)(d+4)}{(\eta + u)^3}.$$

Substitute this into (32):

$$\mathcal{K}'(u) \leq 16 d(d+2)(d+4) \frac{u^2}{\eta^2(\eta + u)}.$$

Integrate from 0 to u . By Lemma 3, $\mathcal{K}(s) \rightarrow \mathcal{K}(0) = 0$ as $s \downarrow 0$, so the fundamental theorem of calculus gives

$$\mathcal{K}(u) \leq 16 d(d+2)(d+4) \int_0^u \frac{s^2}{\eta^2(\eta + s)} ds.$$

A direct calculation gives

$$\int_0^u \frac{s^2}{\eta^2(\eta + s)} ds = \frac{1}{2} \left(\frac{u}{\eta} \right)^2 - \frac{u}{\eta} + \log \left(1 + \frac{u}{\eta} \right) = \Psi \left(\frac{u}{\eta} \right),$$

which proves (27).

For the refinement, use (10) instead. Since $Y_u = U + \sqrt{\eta + u} \tilde{Z}$,

$$\mathbb{E}[\|\nabla^3 \ell_u(Y_u)\|_{\text{F}}^2] \leq \frac{64 \mathbb{E}\|U\|^6}{(\eta + u)^6}.$$

Substituting into (32) gives

$$\mathcal{K}'(u) \leq 16 \mathbb{E}\|U\|^6 \frac{u^2}{\eta^2(\eta + u)^4}.$$

Integrating from 0 to u ,

$$\mathcal{K}(u) \leq 16 \mathbb{E}\|U\|^6 \int_0^u \frac{s^2}{\eta^2(\eta + s)^4} ds.$$

Another direct calculation yields

$$\int_0^u \frac{s^2}{\eta^2(\eta + s)^4} ds = \frac{u^3}{3\eta^3(\eta + u)^3},$$

which proves (28).

Finally, define

$$f(r) := \frac{r^3}{3} - \Psi(r).$$

Then $f(0) = 0$, and

$$f'(r) = r^2 - r + 1 - \frac{1}{1+r} = \frac{r^3}{1+r} \geq 0 \quad (r \geq 0).$$

Therefore $f(r) \geq 0$ for all $r \geq 0$, which proves (29). The bound (30) follows immediately when $u \leq \eta$. $\square$

Remark 2. *Theorem 2 is the a quantification of the “small-step reverse is approximately Gaussian” intuition. The true small parameter is u/η : the new noise added at this step must be small relative to the smoothing already present. This is why the first reverse step is fundamentally different from the later ones.*

5 Application to the diffusion chain

We now specialize the generic result to the DDPM forward chain (1). The idea is the following: at each step $t \geq 2$, the signal entering the step already contains Gaussian noise accumulated from all previous steps. This accumulated noise plays the role of the “pre-existing smoothing” η in the generic setup, and the new noise β_t plays the role of u .

5.1 The bulk steps $t \geq 2$

Define for each $t \geq 2$ the rescaled previous state

$$S_t := \sqrt{1 - \beta_t} X_{t-1}.$$

Then

$$X_t = S_t + \sqrt{\beta_t} Z_t.$$

Because

$$X_{t-1} = \sqrt{\bar{\alpha}_{t-1}} X_0 + \sqrt{1 - \bar{\alpha}_{t-1}} W, \quad W \sim \mathcal{N}(0, I_d),$$

we also have

$$S_t = \sqrt{\bar{\alpha}_t} X_0 + \sqrt{\eta_t} W, \quad \eta_t := (1 - \beta_t)(1 - \bar{\alpha}_{t-1}). \quad (33)$$

Thus the signal entering step t already contains Gaussian smoothing of variance η_t .

Theorem 3 (Bulk one-step Gaussianization bound). *For each $t \geq 2$, define*

$$\mathcal{K}_t := \mathbb{E} \left[D_{\text{KL}}(q_t(\cdot \mid X_t) \parallel g_t^*(\cdot \mid X_t)) \right].$$

Then

$$\mathcal{K}_t \leq 16 d(d+2)(d+4) \Psi\left(\frac{\beta_t}{\eta_t}\right) = 16 d(d+2)(d+4) \Psi\left(\frac{\beta_t}{(1 - \beta_t)(1 - \bar{\alpha}_{t-1})}\right). \quad (34)$$

If, in addition, $\mathbb{E}\|X_0\|^6 < \infty$, then

$$\mathcal{K}_t \leq \frac{16}{3} \frac{\bar{\alpha}_t^3 \beta_t^3}{\eta_t^3 (1 - \bar{\alpha}_t)^3} \mathbb{E}\|X_0\|^6. \quad (35)$$

Finally, if $\beta_t \leq \eta_t$, then

$$\mathcal{K}_t \leq \frac{16}{3} d(d+2)(d+4) \left(\frac{\beta_t}{\eta_t} \right)^3. \quad (36)$$

Proof. By (33), the rescaled variable S_t and the observation $X_t = S_t + \sqrt{\beta_t} Z_t$ fit the generic setup (11) with

$$U = \sqrt{\bar{\alpha}_t} X_0, \quad \eta = \eta_t, \quad u = \beta_t.$$

Applying Theorem 2 gives

$$\mathbb{E} \left[D_{\text{KL}}(\mathcal{L}(S_t | X_t) \parallel \hat{g}_t^*(\cdot | X_t)) \right] \leq 16 d(d+2)(d+4) \Psi \left(\frac{\beta_t}{\eta_t} \right), \quad (37)$$

where $\mathcal{L}(S_t | X_t)$ denotes the conditional distribution of S_t given X_t , and $\hat{g}_t^*(\cdot | X_t)$ is its moment-matched Gaussian approximation.

We would like to replace the left-hand side of (37) with $\mathcal{K}_t$. The map $\varphi(x) = \sqrt{1 - \beta_t} x$ is an invertible affine bijection whose push-forward sends $\mathcal{L}(X_{t-1} | X_t)$ to $\mathcal{L}(S_t | X_t)$. Because affine maps preserve the first two moments, it also sends $g_t^*(\cdot | X_t)$ to the moment-matched Gaussian of $\mathcal{L}(S_t | X_t)$, namely $\hat{g}_t^*(\cdot | X_t)$. By the invariance of KL under invertible affine maps,

$$D_{\text{KL}}(q_t(\cdot | X_t) \parallel g_t^*(\cdot | X_t)) = D_{\text{KL}}(\mathcal{L}(S_t | X_t) \parallel \hat{g}_t^*(\cdot | X_t)).$$

Taking expectations identifies the left-hand side of (37) with $\mathcal{K}_t$, which proves (34). The refined bound (35) and the cubic bound (36) follow by the same argument, starting from (28) and (30) in place of (27); for the refined bound, use $\mathbb{E}\|U\|^6 = \bar{\alpha}_t^3 \mathbb{E}\|X_0\|^6$ and $\eta_t + \beta_t = 1 - \bar{\alpha}_t$ to match the stated form. $\square$

Remark 3 (Why the first step is special). *For $t = 1$, the quantity $\eta_t = (1 - \beta_t)(1 - \bar{\alpha}_{t-1})$ is zero, because $\bar{\alpha}_0 = 1$. In other words, before the first diffusion step there is no pre-existing Gaussian smoothing at all. In this regime, boundary-layer effects can survive and a uniform Gaussian approximation need not hold. The theorem therefore begins at $t = 2$.*

Remark 4 (Bounding the first step under additional smoothness). *If P_0 is itself a Gaussian convolution (that is, $X_0 = U_0 + \sqrt{\eta_0} W_0$ for some $\eta_0 > 0$ and $W_0 \sim \mathcal{N}(0, I_d)$ independent of U_0), then Theorem 2 applies to $t = 1$ as well, giving*

$$\mathcal{K}_1 \leq 16 d(d+2)(d+4) \Psi \left(\frac{\beta_1}{(1 - \beta_1)\eta_0} \right).$$

For general P_0 , the first reverse step requires a different technique; the averaged-KL approach exploits pre-existing smoothing that is simply absent before the first forward step.

Remark 5 (Parallel with practice). *The final reverse step is also treated separately in practice. [3] and [4] write the variational lower bound as*

$$L = L_T + \sum_{t \geq 2} L_{t-1} + L_0, \quad L_0 := -\log p_\theta(X_0 | X_1).$$

The L_{t-1} for $t \geq 2$ are KL divergences between Gaussians and are evaluated in closed form. The term L_0 is different: $p_\theta(X_0 | X_1)$ is a discretized decoder that integrates a continuous Gaussian density over the 256 quantization bins for each pixel.

The reason is the same in both cases. For image data P_0 is supported either on a low-dimensional manifold or on the discrete pixel lattice, so $\mathcal{L}(X_0 | X_1)$ is not a smooth density, and a continuous-Gaussian approximation is qualitatively wrong at $t = 1$. Our bound $A_u \succeq \frac{\eta}{\eta+u} I_d$ degenerates at $\eta = 0$ for the same reason that DDPM implementations need a quantization-aware decoder at the final step.

5.2 The terminal term

The terminal term is easier to bound and follows from the contraction properties of the Ornstein–Uhlenbeck semigroup.

Proposition 3 (Terminal KL contraction). *For any initial law P_0 ,*

$$D_{\text{KL}}(P_T \parallel \mathcal{N}(0, I_d)) \leq \bar{\alpha}_T D_{\text{KL}}(P_0 \parallel \mathcal{N}(0, I_d)).$$

Proof. Write $\gamma := \mathcal{N}(0, I_d)$ throughout this proof. We first establish a one-step contraction. Let μ be any probability measure on $\mathbb{R}^d$, let $X \sim \mu$, and set

$$Y = \sqrt{1 - \beta} X + \sqrt{\beta} Z, \quad Z \sim \gamma \text{ independent of } X.$$

Writing $\tau = -\frac{1}{2} \log(1 - \beta)$,

$$Y = e^{-\tau} X + \sqrt{1 - e^{-2\tau}} Z,$$

which is exactly one step of the Ornstein–Uhlenbeck semigroup at time τ started at μ . Let $(\mu_s)_{s \geq 0}$ denote the resulting OU trajectory, so that $\mu_0 = \mu$ and $\mu_\tau = \mathcal{L}(Y)$. Then

$$\frac{d}{ds} D_{\text{KL}}(\mu_s \parallel \gamma) = -I(\mu_s \parallel \gamma),$$

where $I(\cdot \parallel \gamma)$ is the relative Fisher information. The Gaussian log-Sobolev inequality [1, Prop. 5.5.1] says

$$I(\mu_s \parallel \gamma) \geq 2D_{\text{KL}}(\mu_s \parallel \gamma),$$

so

$$\frac{d}{ds} D_{\text{KL}}(\mu_s \parallel \gamma) \leq -2D_{\text{KL}}(\mu_s \parallel \gamma),$$

and Grönwall’s inequality yields

$$D_{\text{KL}}(\mu_\tau \parallel \gamma) \leq e^{-2\tau} D_{\text{KL}}(\mu \parallel \gamma) = (1 - \beta) D_{\text{KL}}(\mu \parallel \gamma).$$

Applying this contraction to each forward step of the diffusion chain, with $\mu = P_{t-1}$, $\mu_\tau = P_t$, and $\beta = \beta_t$, gives $D_{\text{KL}}(P_t \parallel \gamma) \leq (1 - \beta_t) D_{\text{KL}}(P_{t-1} \parallel \gamma)$. Iterating from $t = 1$ to $t = T$,

$$D_{\text{KL}}(P_T \parallel \gamma) \leq \prod_{t=1}^T (1 - \beta_t) D_{\text{KL}}(P_0 \parallel \gamma) = \bar{\alpha}_T D_{\text{KL}}(P_0 \parallel \mathcal{N}(0, I_d)).$$

□

5.3 Global model-error bound

Let

$$\mathcal{K}_1 := \mathbb{E} \left[D_{\text{KL}}(q_1(\cdot \mid X_1) \parallel g_1^*(\cdot \mid X_1)) \right].$$

We leave this first-step term explicit, for the reason explained in Remark 3.

Corollary 3 (Global upper bound). *Assume $D_{\text{KL}}(P_0 \parallel \mathcal{N}(0, I_d)) < \infty$. Then*

$$D_{\text{KL}}(P_0 \parallel \tilde{P}_0) \leq \bar{\alpha}_T D_{\text{KL}}(P_0 \parallel \mathcal{N}(0, I_d)) + \mathcal{K}_1 + 16 d(d+2)(d+4) \sum_{t=2}^T \Psi \left(\frac{\beta_t}{(1 - \beta_t)(1 - \bar{\alpha}_{t-1})} \right).$$

If, in addition, $\mathbb{E}\|X_0\|^6 < \infty$, then

$$D_{\text{KL}}(P_0 \parallel \tilde{P}_0) \leq \bar{\alpha}_T D_{\text{KL}}(P_0 \parallel \mathcal{N}(0, I_d)) + \mathcal{K}_1 + \frac{16}{3} \mathbb{E}\|X_0\|^6 \sum_{t=2}^T \frac{\bar{\alpha}_t^3 \beta_t^3}{\eta_t^3 (1 - \bar{\alpha}_t)^3}.$$

Finally, if $\beta_t \leq \eta_t$ for every $t \geq 2$, then

$$D_{\text{KL}}(P_0 \parallel \tilde{P}_0) \leq \bar{\alpha}_T D_{\text{KL}}(P_0 \parallel \mathcal{N}(0, I_d)) + \mathcal{K}_1 + \frac{16}{3} d(d+2)(d+4) \sum_{t=2}^T \left(\frac{\beta_t}{\eta_t} \right)^3.$$

Proof. Insert Theorem 3 and Proposition 3 into the pathwise decomposition (3), and then use (2). The refined and cubic forms follow in the same way from (35) and (36). $\square$

6 Schedule optimization

Corollary 3 splits the global model error into three pieces: the first-step term $\mathcal{K}_1$, the terminal contraction $\bar{\alpha}_T D_{\text{KL}}(P_0 \parallel \mathcal{N}(0, I_d))$, and the bulk sum $\sum_{t=2}^T \Psi(\beta_t/\eta_t)$. The schedule affects all three. However, the first two depend on the schedule only through the scalars β_1 and $\bar{\alpha}_T$, and they involve unknown quantities: $\mathcal{K}_1$ is a black box (our argument does not apply at $t = 1$, see Remark 3), and $D_{\text{KL}}(P_0 \parallel \mathcal{N}(0, I_d))$ is an unknown property of the data. We therefore hold β_1 and $\bar{\alpha}_T$ fixed as design parameters — making the two boundary terms constants — and optimize the bulk sum over the remaining degrees of freedom in the schedule.

The optimization problem. Let $\beta_1^* \in (0, 1)$ be the prescribed first-step noise and $\bar{\alpha}_T^* \in (0, 1)$ the prescribed terminal signal level. Choose $\beta_1, \beta_2, \dots, \beta_T \in (0, 1)$ to minimize

$$\sum_{t=2}^T \Psi\left(\frac{\beta_t}{(1 - \beta_t)(1 - \bar{\alpha}_{t-1})}\right) \quad (38)$$

subject to

$$\beta_1 = \beta_1^* \quad \text{and} \quad \bar{\alpha}_T = \bar{\alpha}_T^*,$$

where $\bar{\alpha}_t := \prod_{i=1}^t (1 - \beta_i)$ as in (1). Note that the factor $16 d(d+2)(d+4)$ from Corollary 3 is a constant in this problem, so we omit it throughout.

The obstacle is that the constraint $\prod_{t=1}^T (1 - \beta_t) = \bar{\alpha}_T^*$ does not decouple nicely in the β_t variables, and the objective is a nontrivial function of each β_t . Both difficulties disappear after a change of variables.

6.1 Change of variables

Define the odds variable

$$z_t := \frac{1 - \bar{\alpha}_t}{\bar{\alpha}_t} \quad (t = 0, 1, \dots, T).$$

Then $z_0 = 0$, and for every $t \geq 1$,

$$\bar{\alpha}_t = \frac{1}{1 + z_t}.$$

For the steps $t \geq 2$, also define

$$r_t := \frac{\beta_t}{\eta_t} = \frac{\beta_t}{(1 - \beta_t)(1 - \bar{\alpha}_{t-1})}. \quad (39)$$

The next lemma is the key algebraic relation.

Lemma 4 (Multiplicative dynamics in the odds variable). *For every $t \geq 2$,*

$$1 + r_t = \frac{z_t}{z_{t-1}}. \quad (40)$$

Equivalently,

$$z_t = (1 + r_t)z_{t-1}. \quad (41)$$

Proof. Since $\bar{\alpha}_t = (1 - \beta_t)\bar{\alpha}_{t-1}$, we have

$$\frac{\beta_t}{1 - \beta_t} = \frac{\bar{\alpha}_{t-1} - \bar{\alpha}_t}{\bar{\alpha}_t}.$$

Divide both sides by $1 - \bar{\alpha}_{t-1}$:

$$r_t = \frac{\bar{\alpha}_{t-1} - \bar{\alpha}_t}{\bar{\alpha}_t(1 - \bar{\alpha}_{t-1})}.$$

Now

$$\frac{z_t}{z_{t-1}} = \frac{(1 - \bar{\alpha}_t)/\bar{\alpha}_t}{(1 - \bar{\alpha}_{t-1})/\bar{\alpha}_{t-1}} = \frac{\bar{\alpha}_{t-1}(1 - \bar{\alpha}_t)}{\bar{\alpha}_t(1 - \bar{\alpha}_{t-1})} = 1 + \frac{\bar{\alpha}_{t-1} - \bar{\alpha}_t}{\bar{\alpha}_t(1 - \bar{\alpha}_{t-1})} = 1 + r_t.$$

This proves the lemma. $\square$

6.2 Optimizing the schedule

In the odds coordinate, the constraints $\beta_1 = \beta_1^*$ and $\bar{\alpha}_T = \bar{\alpha}_T^*$ translate to fixing

$$z_1 = \frac{\beta_1^*}{1 - \beta_1^*} \quad \text{and} \quad z_T = \frac{1 - \bar{\alpha}_T^*}{\bar{\alpha}_T^*},$$

and the bulk objective (38) becomes

$$\mathcal{S}_{\text{bulk}} := \sum_{t=2}^T \Psi(r_t). \quad (42)$$

Theorem 4. *Fix $z_1 > 0$ and $z_T > z_1$. Among all schedules with these endpoints, the expression (42) is minimized when*

$$z_t = z_1 q^{t-1}, \quad q := \left(\frac{z_T}{z_1} \right)^{1/(T-1)}, \quad t = 1, \dots, T. \quad (43)$$

Equivalently,

$$r_2 = r_3 = \dots = r_T = q - 1.$$

In terms of the original schedule,

$$\beta_t = \frac{(q - 1)z_{t-1}}{1 + q z_{t-1}}, \quad t = 2, \dots, T. \quad (44)$$

Proof. The constraint $\beta_1 = \beta_1^*$ fixes z_1 , leaving $r_2, \dots, r_T$ as the free variables. Iterating the recursion (41) telescopes to

$$z_T = z_1 \prod_{t=2}^T (1 + r_t),$$

so the remaining constraint $\bar{\alpha}_T = \bar{\alpha}_T^*$ — equivalently, z_T fixed to its prescribed value — becomes the single additive condition

$$\sum_{t=2}^T \log(1 + r_t) = \log\left(\frac{z_T}{z_1}\right).$$

The optimization problem is therefore:

$$\min_{r_t \geq 0} \sum_{t=2}^T \Psi(r_t) \quad \text{subject to} \quad \sum_{t=2}^T \log(1 + r_t) = \log\left(\frac{z_T}{z_1}\right).$$

Introduce the variable

$$s_t := \log(1 + r_t) \quad (t = 2, \dots, T).$$

Then $r_t = e^{s_t} - 1$, the constraint becomes

$$\sum_{t=2}^T s_t = \log\left(\frac{z_T}{z_1}\right),$$

and the objective becomes

$$\sum_{t=2}^T F(s_t), \quad F(s) := \Psi(e^s - 1).$$

A direct calculation gives

$$F'(s) = (e^s - 1)^2, \quad F''(s) = 2e^s(e^s - 1) \geq 0 \quad (s \geq 0).$$

Thus F is convex on $[0, \infty)$. Jensen's inequality therefore implies that $\sum_{t=2}^T F(s_t)$ is minimized when all the s_t are equal:

$$s_2 = s_3 = \dots = s_T = \frac{1}{T-1} \log\left(\frac{z_T}{z_1}\right) = \log q.$$

Hence all the $r_t = e^{s_t} - 1$ are equal to $q - 1$. Iterating (41) gives

$$z_t = z_1 q^{t-1},$$

which is (43). Finally, solve (39) for β_t :

$$\beta_t = r_t(1 - \beta_t)(1 - \bar{\alpha}_{t-1}) = r_t(1 - \beta_t) \frac{z_{t-1}}{1 + z_{t-1}}.$$

Solving for β_t gives

$$\beta_t = \frac{r_t z_{t-1}}{1 + z_{t-1} + r_t z_{t-1}} = \frac{r_t z_{t-1}}{1 + (1 + r_t) z_{t-1}}.$$

Substituting $r_t = q - 1$ and $1 + r_t = q$ proves (44). □

The complete recipe. Collecting the pieces, Theorem 4 gives an explicit, closed-form prescription for the entire schedule. The inputs are the number of reverse steps T and two boundary parameters: the first-step noise $\beta_1 \in (0, 1)$ and the terminal signal level $\bar{\alpha}_T \in (0, 1)$. The output is the full schedule $\beta_1, \beta_2, \dots, \beta_T$, computed in three lines:

$$1. \text{ Odds endpoints. } z_1 = \frac{\beta_1}{1 - \beta_1}, \quad z_T = \frac{1 - \bar{\alpha}_T}{\bar{\alpha}_T}.$$

2. **Geometric ratio.** $q = (z_T/z_1)^{1/(T-1)}$.

3. **Schedule.** $z_t = z_1 q^{t-1}$, and for $t = 2, \dots, T$, $\beta_t = \frac{(q-1)z_{t-1}}{1+qz_{t-1}}$.

This is the main practical takeaway of the paper.

In practice, β_1 and $\bar{\alpha}_T$ can be inherited from existing conventions in the diffusion-model literature. For instance, the DDPM linear schedule [3] uses $\beta_1 = 10^{-4}$ and $T = 1000$, which induces $\bar{\alpha}_T \approx 4 \times 10^{-5}$; substituting these into the three steps above gives a concrete schedule that uses the same computational budget and the same endpoints as DDPM, but re-distributes the noise optimally across the reverse steps. Figure 2 shows the result.

Remark 6 (Endpoint selection). *For a given number of steps T , the geometric-odds schedule is fully determined by the two endpoint parameters $(\beta_1^*, \bar{\alpha}_T^*)$. In general, these can be tuned on a validation set with a two-dimensional grid search — a substantial reduction from the T -dimensional search over arbitrary schedules.*

The search space can be reduced further to a single parameter by imposing the midpoint constraint $\bar{\alpha}_{[T/2]} = \frac{1}{2}$. This is a natural condition: it asks that the signal and noise contributions have equal variance at the temporal midpoint of the chain. Under the geometric-odds schedule, $\bar{\alpha}_t = 1/(1+z_t)$, so $\bar{\alpha}_{[T/2]} = \frac{1}{2}$ is equivalent to $z_{[T/2]} = 1$, which for large T reduces to $z_1 z_T = 1$, or equivalently $\bar{\alpha}_T \approx \beta_1$. The schedule is then determined by β_1 alone: β_1 controls the noise at the signal end, and the noise end is fixed by symmetry.

Remark 7. *If one uses the cubic approximation $\Psi(r) \approx r^3/3$ valid for $r \leq 1$, the same geometric schedule remains optimal. Indeed, after writing $r_t = e^{s_t} - 1$, the objective becomes*

$$\sum_{t=2}^T (e^{s_t} - 1)^3,$$

and the function $s \mapsto (e^s - 1)^3$ is also convex on $[0, \infty)$.

6.3 What if the prefactor were t -dependent?

The optimization above treats $16d(d+2)(d+4)$ as a t -independent constant, which is correct under the universal third-derivative bound (9) but understates the true scaling. The refined bound (35) already exhibits a genuine t -dependence:

$$\mathcal{K}_t \leq \frac{16}{3} \mathbb{E} \|X_0\|^6 \frac{\bar{\alpha}_t^3 \beta_t^3}{\eta_t^3 (1 - \bar{\alpha}_t)^3} = \frac{16}{3} \mathbb{E} \|X_0\|^6 \frac{r_t^3}{z_t^3}, \quad (45)$$

since $\bar{\alpha}_t/(1 - \bar{\alpha}_t) = 1/z_t$. The prefactor $1/z_t^3$ decays sharply as the chain progresses: at the first bulk step z_1 is small (large prefactor), while at the terminal step z_T is large (vanishing prefactor). This is not a quirk of the algebra but a real feature: as the signal is smoothed, its contribution to $\nabla^3 \log p_t$ decays, and the per-step Gaussianization error decays with it.

What the right $K(t)$ should look like. Both bounds are instances of the exact identity $\sigma^6 \nabla^3 \ell(Y) = \kappa_3(V | Y)$ (Corollary 1), and they differ in how they bound $\mathbb{E} \|\kappa_3(V | Y)\|^2$:

- The universal bound replaces V by the noise Z and uses $\mathbb{E} \|Z\|^6 = d(d+2)(d+4)$.
- The refined bound uses $\mathbb{E} \|V\|^6$ directly.

Neither is sharp. The structurally correct envelope is

$$\mathbb{E}\|\nabla^3 \ell_t\|_F^2 \lesssim \frac{\bar{\alpha}_t^3 \|\kappa_3(X_0)\|_F^2}{(1 - \bar{\alpha}_t)^6}, \quad (46)$$

obtained by tracking the third cumulant of the marginal X_t , which scales as $\bar{\alpha}_t^{3/2} \kappa_3(X_0)$. For Gaussian data, $\|\kappa_3(X_0)\|_F = 0$ and the bound vanishes identically, recovering the fact that the reverse process for Gaussian data is exactly Gaussian. The two existing bounds are loose by exactly the factor that distinguishes $\mathbb{E}\|X_0\|^6$ (or $d(d+2)(d+4)$) from the third-cumulant norm $\|\kappa_3(X_0)\|_F^2$ — a quantitative measure of non-Gaussianity. We therefore propose the unified envelope

$$\mathcal{K}_t \leq \min \left\{ 16 d(d+2)(d+4) \Psi(r_t), \frac{c_3 \|\kappa_3(X_0)\|_F^2 r_t^3}{z_t^3} \right\} \quad (47)$$

for an absolute constant $c_3 > 0$, with the universal piece tight when $\bar{\alpha}_t$ is close to 1 and the cumulant piece tight when $\bar{\alpha}_t$ is small. The crossover sits near

$$z_t = \left(\frac{\|\kappa_3(X_0)\|_F^2}{d(d+2)(d+4)} \right)^{1/3},$$

which for normalized data with $\mathbb{E}\|X_0\|^2 = O(d)$ is typically $O(1)$, i.e., near the temporal midpoint of the chain.

The optimizer under the cumulant prefactor. If we drop the universal piece and optimize the cumulant bound alone, the bulk objective becomes

$$\sum_{t=2}^T \frac{r_t^3}{z_t^3} = \sum_{t=2}^T \frac{\bar{\alpha}_t^3 \beta_t^3}{\eta_t^3 (1 - \bar{\alpha}_t)^3}.$$

Since z_t depends on $r_2, \dots, r_t$ through $z_t = z_1 \prod_{i \leq t} (1 + r_i)$, the problem is no longer separable in r_t alone. It is, however, separable in the variable $\bar{\alpha}_t$ directly: in the small- β_t regime $\eta_t \approx 1 - \bar{\alpha}_t \approx 1 - \bar{\alpha}_{t-1}$, and (45) reduces to the per-step cost

$$\mathcal{K}_t \lesssim \bar{\alpha}_{t-1}^3 \beta_t^3. \quad (48)$$

Equalizing this across steps gives $\beta_t \propto 1/\bar{\alpha}_{t-1}$, which integrates to the *linear-decay* schedule

$$\bar{\alpha}_t = 1 - \frac{(1 - \bar{\alpha}_T^*) t}{T}, \quad \beta_t \approx \frac{1}{T - t + 1}. \quad (49)$$

Under this schedule the per-step cost is $\bar{\alpha}_{t-1}^3 \beta_t^3 \equiv 1/T^3$, so the total bulk error is $O(1/T^2)$ — the same order as the geometric schedule, but with a smaller absolute constant whenever $\|\kappa_3(X_0)\|_F^2$ is appreciably less than $d(d+2)(d+4) \mathbb{E}\|X_0\|^{-6} \cdot \mathbb{E}\|X_0\|^6$, i.e., whenever the data is meaningfully closer to Gaussian than its sixth moment alone would suggest.

Why we adopt the universal bound for the main result. The geometric-odds schedule (43) is the right answer to the data-free version of the optimization problem; the linear-decay schedule (49) is the right answer when one is willing to commit to a model of the data’s third cumulant. We default to the universal version because (i) it is data-free; (ii) it yields a clean closed form determined by two interpretable scalars; and (iii) the resulting bound holds uniformly over all

data with finite KL divergence to a Gaussian, including heavy-tailed and strongly non-Gaussian distributions for which $\|\kappa_3(X_0)\|_F^2$ need not be the dominant scale. A practical hybrid would use (47) and take steps geometrically in z_t early in the chain (where the universal piece is tight) and linearly in $\bar{\alpha}_t$ late in the chain (where the cumulant piece is tight), with a crossover near the temporal midpoint.

Figure 1 illustrates one such hybrid, constructed by stitching the geometric-odds segment (steps 1 through $T/2$, targeting $\bar{\alpha}_{T/2} = \frac{1}{2}$) to the linear-decay segment (steps $T/2$ through T , interpolating linearly from $\bar{\alpha}_{T/2} = \frac{1}{2}$ to $\bar{\alpha}_T^*$). Panels (a)–(b) show that this construction is smooth at the crossover: $\bar{\alpha}_t$ is continuous and monotone, and z_t transitions from a straight line on the log scale (geometric segment) to the curve produced by linear-in- $\bar{\alpha}_t$ decay (right segment). Panels (c)–(d) confirm the design intent: under the universal cost $\Psi(r_t)$ the hybrid is flat across the first half (matching the geometric optimum there), and under the cumulant cost $\bar{\alpha}_{t-1}^3 \beta_t^3$ it is flat across the second half (matching the linear-decay optimum there). Neither pure schedule has both properties: the pure geometric explodes the cumulant cost near $t = T$, while the pure linear-decay explodes the universal cost near $t = 2$.

6.4 Comparison of schedules

Figure 2 compares our geometric-odds schedule to the DDPM linear schedule [3] and the cosine schedule [4], all with $T = 1000$ steps. The two baseline schedules have different endpoints — DDPM linear uses $\beta_1 = 10^{-4}$ and reaches $\bar{\alpha}_T \approx 4 \times 10^{-5}$, while the cosine schedule (with its standard β_t -clipping at 0.999) uses $\beta_1 \approx 4 \times 10^{-5}$ and drives $\bar{\alpha}_T$ to $\approx 2 \times 10^{-9}$ — so we plot two instances of our schedule, one matched to each set of endpoints. Panel (b) shows that both versions of our schedule make z_t grow geometrically (a straight line on the log scale), as (43) predicts. Panel (c) shows that the noise-to-smoothing ratio r_t is constant under each of our schedules, while it varies by several orders of magnitude under both baselines. Panel (d) shows the resulting per-step cost $\Psi(r_t)$: our schedules distribute the cost uniformly across steps, whereas DDPM linear and cosine concentrate it on a handful of expensive endpoint steps.

7 Empirical evaluation

We evaluate the geometric-odds schedule of Theorem 4 on four image datasets, comparing it head-to-head with the two standard baselines (DDPM linear and cosine).

7.1 Setup

Datasets. MNIST, Fashion-MNIST, and CIFAR-10 are evaluated at 32×32 (60K/60K/50K training images, 10K test images each). ImageNet-64 is evaluated at 64×64 (≈ 1.28 M training images, 50K validation images).

Architecture and training. All experiments use the U-Net of [4] with 128 base channels, three residual blocks per resolution, and attention at resolutions 16 and 8; the channel multipliers are (1, 2, 2, 2) for the 32×32 datasets and (1, 2, 3, 4) for ImageNet-64. Models are trained with $T = 4000$ forward steps, learning rate 10^{-4} , batch size 128, and EMA rate 0.9999. CIFAR-10, Fashion-MNIST, and MNIST are trained for 500K iterations; ImageNet-64 for 200K iterations. Sampling uses the full $T = 4000$ -step reverse chain.

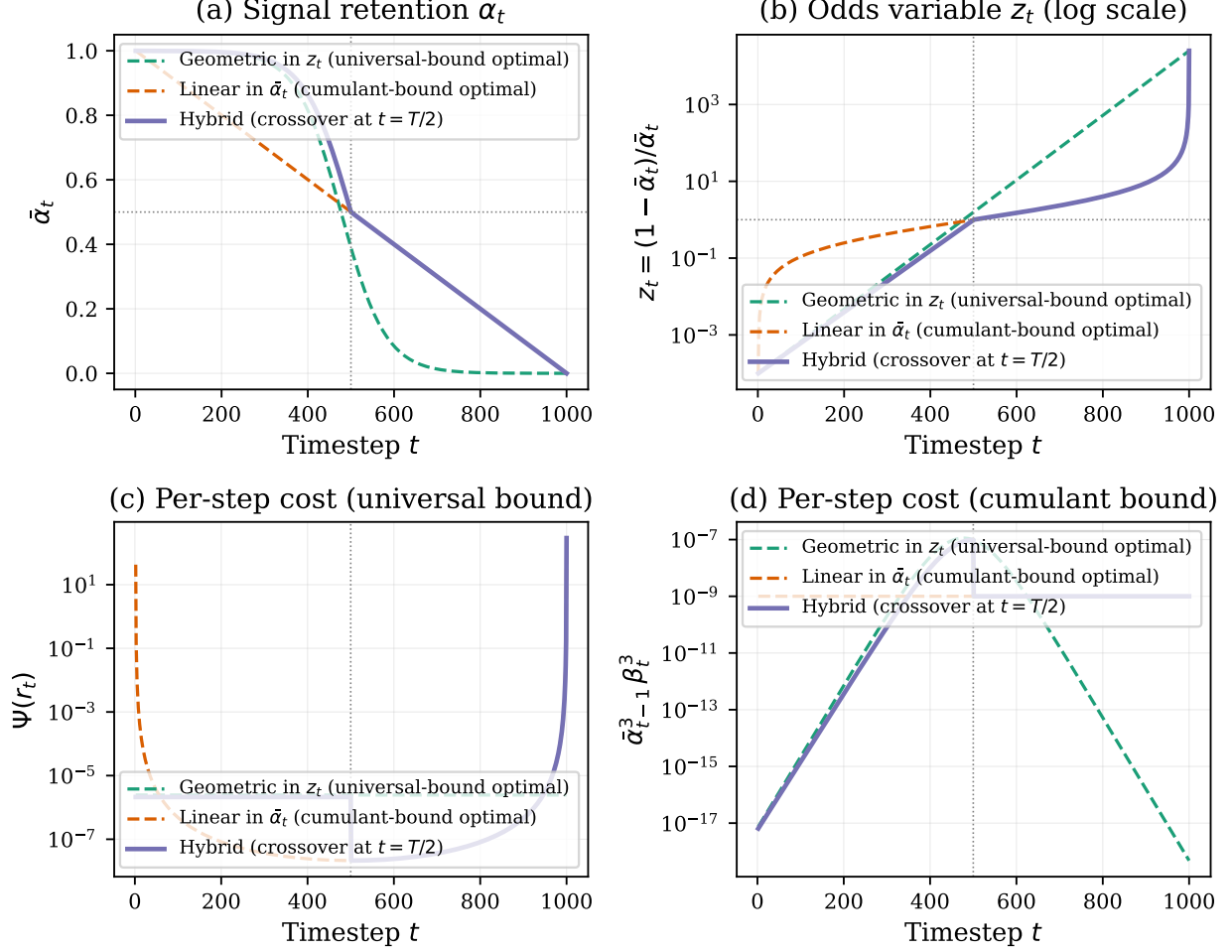


Figure 1: A hybrid schedule that interpolates between the two regimes. The hybrid (solid purple) is geometric in z_t for $t \leq T/2$ and linear in $\bar{\alpha}_t$ for $t \geq T/2$, stitched at the natural midpoint $\bar{\alpha}_{T/2} = \frac{1}{2}$. Endpoints are $\beta_1 = 10^{-4}$ and $\bar{\alpha}_T = 4 \times 10^{-5}$; the dotted gray lines mark the crossover. **(a)** Signal retention $\bar{\alpha}_t$. **(b)** Odds variable z_t . **(c)** Per-step cost $\Psi(r_t)$ under the universal bound: the hybrid matches the geometric optimum in the first half. **(d)** Per-step cost $\bar{\alpha}_{t-1}^3 \beta_t^3$ under the cumulant bound: the hybrid matches the linear-decay optimum in the second half. Each pure schedule equalizes only one of the two costs; the hybrid equalizes whichever is tight at each time.

Schedules. We compare four schedules in two groups.

- *Baselines:* **linear** [3] and **cosine** [4].
- *Geometric-odds, matched endpoints:* **geometric-linear** uses the geometric-odds recipe of Theorem 4 with the same endpoints $(\beta_1, \bar{\alpha}_T)$ as the DDPM linear schedule, and **geometric-cosine** uses the same endpoints as the cosine schedule. In both cases any difference from the matched baseline comes only from the schedule shape between the endpoints.

Training objectives. We test three standard objectives from [4]: L_{simple} (MSE on the noise prediction with a fixed variance), L_{hybrid} (MSE plus a learned variance, trained with the rescaled

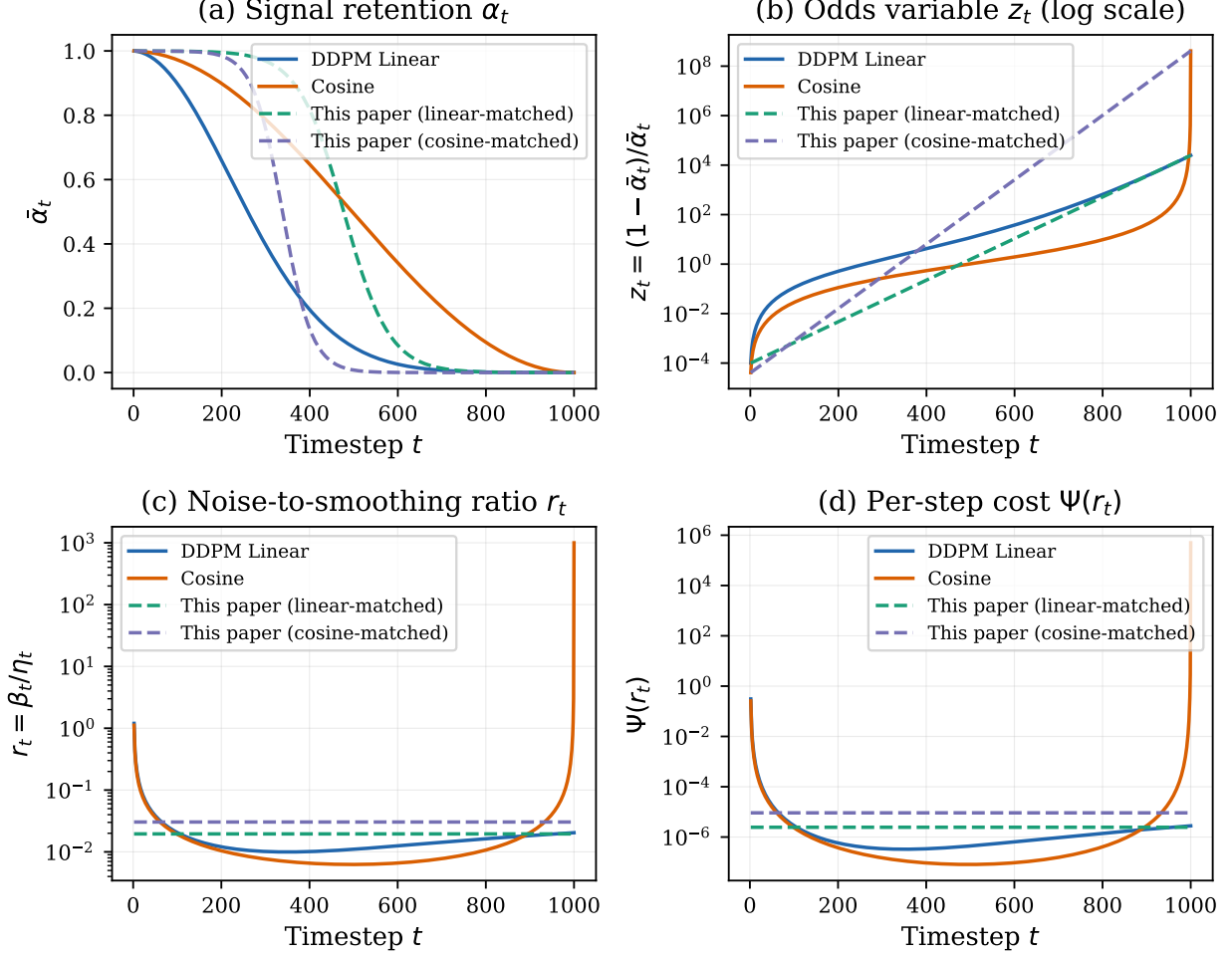


Figure 2: Comparison of noise schedules with $T = 1000$. **(a)** Signal retention $\bar{\alpha}_t$. **(b)** Odds variable z_t on a log scale; our schedule is a straight line in both endpoint configurations. **(c)** Noise-to-smoothing ratio $r_t = \beta_t/\eta_t$; constant under our schedule, and varying by several orders of magnitude under DDPM linear and cosine. **(d)** Per-step cost $\Psi(r_t)$; uniform under our schedule, and sharply concentrated at the endpoints under DDPM linear and cosine. Because the DDPM linear and cosine schedules use different $(\beta_1, \bar{\alpha}_T)$ endpoints, we plot two versions of our schedule, each matched to one of them.

variational lower bound), and L_{vlb} (the full variational lower bound, trained with importance-sampled t). The L_{simple} weighting is implicitly fit to the linear DDPM schedule and is mis-specified for substantially different schedules; L_{hybrid} and L_{vlb} are the well-specified comparisons.

Metrics. We report NLL in bits per dimension on the held-out test set, FID against the training set (50K generated samples for the 32×32 datasets, 10K for ImageNet-64), and total variation (TV) between pixel-intensity histograms of generated and test samples as an auxiliary check on distributional match.

7.2 Matched-endpoint comparison

Table 1 reports NLL and FID for the four {baseline, geometric} schedule pairs across all (dataset, objective) combinations. Within each pair the better value is bolded.

Table 1: Matched-endpoint comparison: holding $(\beta_1, \bar{\alpha}_T)$ fixed and varying only the schedule shape. Bold marks the better of each {baseline, geometric} pair (linear vs. **g-lin**, cosine vs. **g-cos**). NLL in bits/dim; FID against the training set. Lower is better for both metrics.

Dataset	Obj.	NLL (bits/dim)				FID			
		linear	g-lin	cosine	g-cos	linear	g-lin	cosine	g-cos
MNIST	simple	0.960	0.647	0.755	0.344	2.599	0.516	2.173	0.483
MNIST	hybrid	0.388	0.346	0.346	0.286	2.574	0.494	2.258	0.434
MNIST	vlb	0.477	0.391	0.316	0.313	0.580	0.610	0.433	0.450
F-MNIST	simple	1.453	1.746	1.215	0.862	2.436	1.512	2.221	1.438
F-MNIST	hybrid	0.947	0.887	0.828	0.761	2.321	1.547	2.352	1.231
F-MNIST	vlb	1.096	0.922	0.817	0.773	1.278	1.399	1.772	1.254
CIFAR-10	simple	3.418	8.922	3.298	4.875	9.769	6.796	5.550	9.684
CIFAR-10	hybrid	3.296	3.012	3.207	2.921	8.421	7.114	5.475	9.038
CIFAR-10	vlb	3.090	3.009	2.981	2.925	10.524	11.290	11.756	10.694
ImageNet-64	simple	3.522	10.760	3.441	6.068	30.746	35.093	27.780	39.102
ImageNet-64	hybrid	3.398	3.329	3.367	3.316	30.159	34.555	28.799	40.381
ImageNet-64	vlb	3.341	3.327	3.357	3.310	41.213	45.333	62.814	43.775

Three patterns are visible in Table 1. First, for the two well-specified objectives L_{hybrid} and L_{vlb} , the geometric-odds schedule improves NLL on *every* (dataset, objective) cell relative to its matched baseline, with the largest absolute gains on the smaller datasets (e.g., MNIST/hybrid: $0.346 \rightarrow 0.286$ bits/dim under the cosine endpoints; Fashion-MNIST/hybrid: $0.828 \rightarrow 0.761$). On CIFAR-10 the corresponding gains are 0.06–0.29 bits/dim, and on ImageNet-64 they are 0.05–0.07 bits/dim — modest in absolute terms, but consistent across the four matched cells.

Second, on MNIST and Fashion-MNIST the geometric-cosine schedule also dominates the cosine baseline on FID across all three objectives, often by large margins (e.g., MNIST/hybrid FID $2.258 \rightarrow 0.434$; Fashion-MNIST/hybrid FID $2.352 \rightarrow 1.231$). On CIFAR-10 and ImageNet-64 the FID picture is mixed: the geometric schedule typically improves FID under the linear endpoints but not under the cosine endpoints, where the baseline cosine schedule already concentrates much of its noise budget into the high- $\bar{\alpha}_t$ region that contributes most to perceptual quality. This remaining FID gap on the larger datasets is consistent with the cumulant-prefactor refinement of Section 6.3: the universal bound (34) that the geometric schedule minimizes treats every step as worst-case in the data’s third cumulant, whereas FID is sensitive to fine perceptual structure that a data-aware envelope would weight differently.

Third, the L_{simple} rows of the table contain large NLL anomalies, most visibly $L_{\text{simple}}/\text{geometric-linear}$ on CIFAR-10 (8.92 bits/dim) and ImageNet-64 (10.76 bits/dim). These reflect the implicit fit of L_{simple} ’s loss weighting to the DDPM linear schedule: under a substantially different schedule the objective is no longer a tight surrogate for the variational lower bound, and bits/dim degrades even when sample quality is unaffected. The well-posed comparisons are therefore those in the L_{hybrid} and L_{vlb} rows.

7.3 Sample quality

Figure 3 shows samples from the cosine baseline and from the matched geometric-cosine schedule, both trained with L_{hybrid} , on all four datasets. Visual fidelity is comparable on every dataset; on MNIST and Fashion-MNIST the geometric-cosine samples are visibly crisper and contain fewer of the low-contrast smears characteristic of poorly calibrated reverse chains, consistent with the FID improvements those datasets show in Table 1.

7.4 Discussion

The empirical evidence supports the schedule-shape prediction of Theorem 4: holding endpoints fixed, the geometric-odds schedule improves bits/dim on every well-specified cell in Table 1, and improves FID on the smaller datasets. The remaining FID gap on the larger datasets points to the data-dependent refinement of Section 6.3: the universal bound that the geometric schedule minimizes does not see the data’s third cumulant, and the hybrid schedule of Figure 1, which switches between the universal and cumulant-driven regimes near the temporal midpoint, is a natural direction for future empirical work.

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A Technical lemmas

Lemma 5 (Heat expectation identity). *Let p_u be the density of $Y_u = U + \sqrt{\eta + u}W'$, so that p_u satisfies the heat equation $\partial_u p_u = \frac{1}{2}\Delta p_u$. Let $f_u : \mathbb{R}_{>0} \times \mathbb{R}^d \rightarrow \mathbb{R}$ be C^1 in u and C^2 in y , and assume that $f_u, \partial_u f_u, \nabla f_u, \Delta f_u$ all have at most polynomial growth in y , locally uniformly in u . Assume further that U has enough moments that every such polynomial bound is integrable against p_u . Then for every $u > 0$,*

$$\frac{d}{du}\mathbb{E}[f_u(Y_u)] = \mathbb{E}[\partial_u f_u(Y_u)] + \frac{1}{2}\mathbb{E}[\Delta f_u(Y_u)]. \quad (50)$$

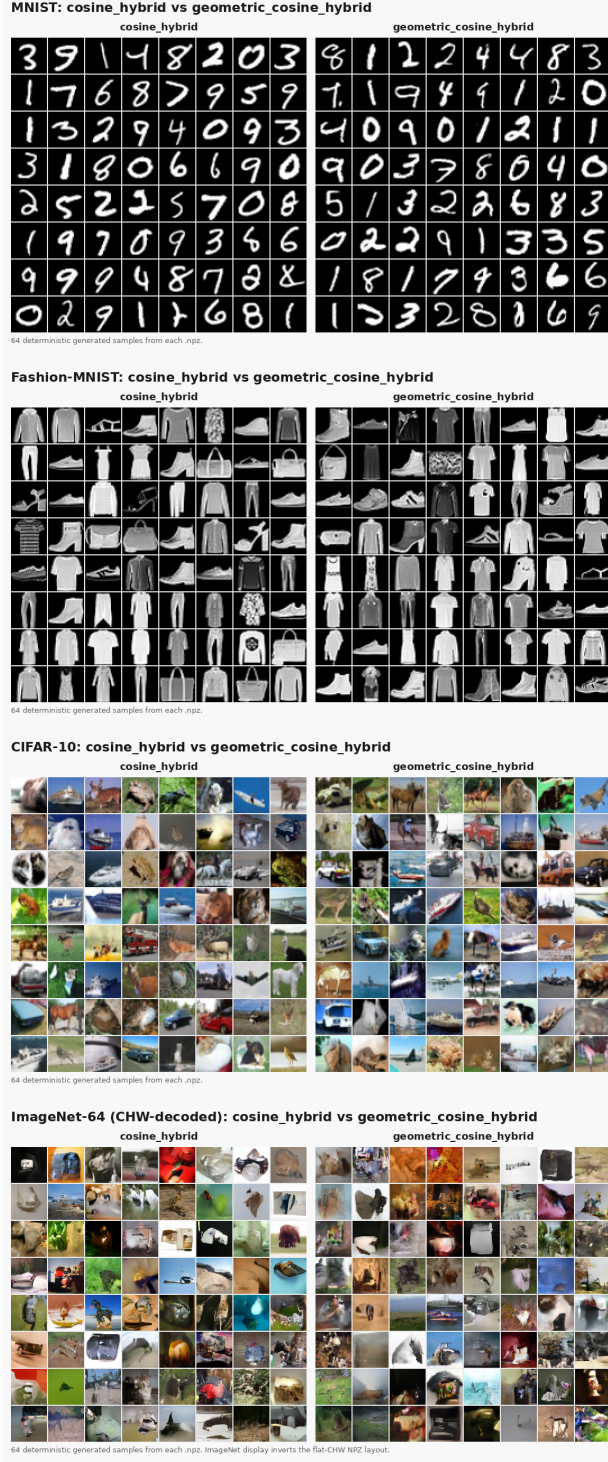


Figure 3: Random samples from `cosine`/ L_{hybrid} (left of each pair) and `geometric-cosine`/ L_{hybrid} (right of each pair) on MNIST, Fashion-MNIST, CIFAR-10, and ImageNet-64. Both schedules share endpoints $(\beta_1, \bar{\alpha}_T)$; only the schedule shape between the endpoints differs.

Proof. Write $\mathbb{E}[f_u(Y_u)] = \int f_u(y) p_u(y) dy$. Under the polynomial-growth and moment hypotheses,

this is a finite number, and differentiation under the integral is justified by dominated convergence (with a dominating function of the form “polynomial in y times p_u ”):

$$\frac{d}{du} \int f_u p_u dy = \int (\partial_u f_u) p_u dy + \int f_u (\partial_u p_u) dy.$$

Using the heat equation $\partial_u p_u = \frac{1}{2} \Delta p_u$, it remains to show

$$\int f_u \Delta p_u dy = \int (\Delta f_u) p_u dy. \quad (51)$$

We prove (51) by a cutoff argument. Fix a smooth radial cutoff $\chi : \mathbb{R}^d \rightarrow [0, 1]$ with $\chi \equiv 1$ on B_1 , $\chi \equiv 0$ outside B_2 , and set $\chi_R(y) := \chi(y/R)$, so that $\chi_R \equiv 1$ on B_R , $\chi_R \equiv 0$ outside B_{2R} , and $\|\nabla \chi_R\|_\infty \leq CR^{-1}$, $\|\Delta \chi_R\|_\infty \leq CR^{-2}$ for a constant C depending only on χ . Because $\chi_R f_u$ has compact support, standard integration by parts on compactly supported functions (with vanishing boundary contribution) gives

$$\int \chi_R f_u \Delta p_u dy = \int p_u \Delta(\chi_R f_u) dy.$$

Expanding $\Delta(\chi_R f_u) = \chi_R \Delta f_u + 2\nabla \chi_R \cdot \nabla f_u + f_u \Delta \chi_R$,

$$\int \chi_R f_u \Delta p_u dy = \int \chi_R (\Delta f_u) p_u dy + 2 \int p_u \nabla \chi_R \cdot \nabla f_u dy + \int p_u f_u \Delta \chi_R dy. \quad (52)$$

We now let $R \rightarrow \infty$. The polynomial-growth and moment hypotheses make each of $|f_u| p_u$, $|f_u \Delta p_u| = 2|f_u \partial_u p_u|$, $|\nabla f_u| p_u$, and $|\Delta f_u| p_u$ integrable (in the second case, because $\partial_u p_u = \frac{1}{2} \Delta p_u$ inherits polynomial-times-Gaussian-derivative decay from the convolution form of p_u). By dominated convergence, the left-hand side of (52) tends to $\int f_u \Delta p_u$ and the first term on the right tends to $\int (\Delta f_u) p_u$. The two cross terms vanish because

$$\left| 2 \int p_u \nabla \chi_R \cdot \nabla f_u dy \right| \leq \frac{2C}{R} \int |\nabla f_u| p_u dy \xrightarrow{R \rightarrow \infty} 0, \quad \left| \int p_u f_u \Delta \chi_R dy \right| \leq \frac{C}{R^2} \int |f_u| p_u dy \xrightarrow{R \rightarrow \infty} 0,$$

using the uniform bounds on $\nabla \chi_R$ and $\Delta \chi_R$ together with finiteness of the remaining integrals. This proves (51), and hence (50). $\square$

Remark 8 (Generality of the technique). *The polynomial-growth hypothesis is chosen because every application of Lemma 5 in this paper is with an f_u of that type; verification is immediate and the hypothesis is easy to state. The cutoff argument in the proof uses nothing specific to polynomial growth, and gives the same identity whenever $|f_u| p_u$, $|f_u \partial_u p_u|$, $|\nabla f_u| p_u$, $|\Delta f_u| p_u$ are all integrable and a locally uniform dominating function exists in the differentiation-under-the-integral step.*